

Sheet 6 - risk

1. EU theorem requires preferences to be

- i. Complete
 - ii. Transitive
 - iii. Continuous
 - iv. Satisfying Independence.
- $\left. \begin{array}{l} \text{i. Complete} \\ \text{ii. Transitive} \end{array} \right\} \text{rational} \left\} \text{representable} \right.$
- in order to have an EU representation.

And $U: \mathcal{L} \rightarrow \mathbb{R}$ has EU form iff it's linear, i.e.

$$U\left(\sum_k q_k L_k\right) = \sum_k q_k U(L_k) \quad \text{for any } L_k \in \mathcal{L}, \sum_k q_k = 1$$

i.e. there exist ^{numbers} ~~probabilities~~ $u(x_i)$ s.t. $U(L) = \sum p_i u(x_i) \quad \forall L \in \mathcal{L}$

a. Clearly not linear in probabilities; it's a quadratic structure.

$U(\alpha L + (1-\alpha)L') \neq \alpha U(L) + (1-\alpha)U(L')$ because the ^{mixture} ~~average~~ will have less dispersion than the weighted average of individual dispersions (by Jensen's inequality properties of var)

b. ~~Also nonlinear. Consider e.g. $L_1 = [1, x_1]$ and $L_2 = [0.5, 0.5, x_1, \frac{x_2}{2}]$.~~

~~Then $U(\frac{3}{4}L_1 + \frac{1}{4}L_2) = \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{2} = \frac{7}{8}$~~

$$U(\alpha L + (1-\alpha)L') = \max\{\alpha p_i + (1-\alpha)p_i'\} \text{ for each outcome } i.$$

But consider e.g. $L = [1; x_1]$; $L' = [1; x_2]$, for $\alpha > \frac{1}{2}$ w.l.o.g.

Utility of mixture is α . But weighted avg. of utilities is 2α .

So not EU-consistent.

c. $U(\alpha L + (1-\alpha)L') = \sum_{i \in \mathcal{I}} (\alpha p_i + (1-\alpha)p_i') = \alpha \sum p_i U(L) + (1-\alpha) \sum p_i' U(L')$
So it has EU form.

d. Not EU. Consider $L = [1; \overset{x_1}{\frac{x_1}{2}}]$, $L' = [1; \overset{x_2}{\frac{x_2}{2}}]$. Then

$$U(\alpha L + (1-\alpha)L') = u(\frac{x_2}{2}) \quad \text{while} \quad \alpha U(L) + (1-\alpha)U(L') = \alpha u(\frac{x_1}{2}) + (1-\alpha)u(\frac{x_2}{2})$$

But these can't be equal, since it's a convex combination of $u(x_1)$ and $u(x_2)$ with $u(x_1) > u(x_2)$, so any mix is also $> u(x_2)$, for $\alpha \in (0, 1)$.

Q1 correct conclusions! linearity is a property of an EU function; ideally would need to check whether underlying preferences are consistent with the EU theorem i.e., check for violations of independence/continuity by finding a suitable counter-

- e) Consider $L = [0.6, 0.4; x_1, x_2]$; $L' = [0.4, 0.6; x_2, x_3]$.
 Then $U(\alpha L + (1-\alpha)L') = 2$, which in general
 $\neq \alpha \cdot 1 + (1-\alpha) \cdot 3$. The mode changes in mixtures.
 So not EU.

2. Suppose Nature has determined that the prize is $\bar{x}$. Then the two options available, from some given agent's perspective, are
 $L_1 = [1/N, 1-1/N; \bar{x}, \omega]$ and $L_2 = [1; \bar{x} + \bar{x}/N]$
 $\omega = u(0) = 0$

So the conditional expected utilities are - wlog setting
 $EU[L_1 | X = \bar{x}] = \frac{1}{N} u(\bar{x})$
 $EU[L_2 | X = \bar{x}] = u(\frac{1}{N} \bar{x})$.

By definition of strict concavity, the chord between ^{any} two points always lies below the function u on that interval. So considering in particular the interval points 0 and $\bar{x}$, we have
 $u(\frac{1}{N} \bar{x}) > \frac{1}{N} u(\bar{x}) + (1-\frac{1}{N}) u(0) = \frac{1}{N} u(\bar{x})$.

Thus $L_1 \succ L_2 | \bar{x}$.

And this holds for all $\bar{x}$. So by law of iterated expectations,
 $L_1 \succ L_2$, i.e. risk sharing strictly better. great

3. PhD is lottery $L_p = [4, 0.8; 0.1, 0.9]$. Not doing it is
 $L_n = [1; 1]$.
 $L_n \sim L_p$.

a) $EU[L_p] = 0.1 \times 1 + 0.9 \times 0 = \underline{0.1} = EU[L_n] = u(1)$.

~~b) The EV of this the PhD lottery $EU[L_p] = 0.1 \times 4 + 0.9 \times 0.8 = 0.76$.
 And $EU[L_n] = 1$.
 So Ann must be risk-seeking because despite L_n having strictly higher EV, she (weakly) prefers the riskier L_p .~~

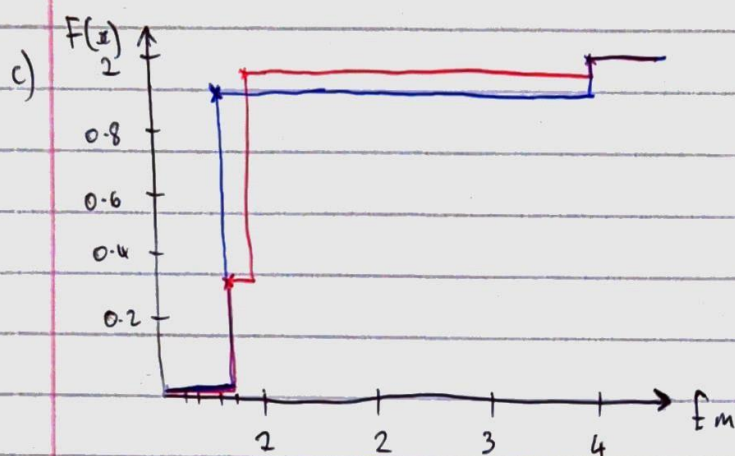
- b) We can't conclude anything about Ann's global risk attitude because we only have her preferences over a single pair of

lotteries. But locally, note that

$$EV[L_P] = 0.1 \times 4 + 0.9 \times 0.8 = 1.12 > EV[L_N] = 1.$$

and she's indifferent between L_P and L_N , i.e. $CE[L_P] = 1 = EV[L_P]$

and $RP(L_P) = 0.12 > 0$. So she's risk-averse in this region.



PhD

None

N doesn't FOSD P
because $P(E_{km} | N) < P(E_{km} | P)$.

However N does SOSD P.

Visually, the CDF crosses P's once from below, so

given that they have the same mean ($EV[L_N] = 1.12$), F_P is a mean-preserving spread of F_N , so all risk-averse agents prefer N.

d) $L_P' = [q, 1-q; 1, 4]$. $L_N' = [0.3, 0.64, 0.06; 0.8, 1, 4]$.

We need $q \cdot u(1) + (1-q) \cdot u(4) = 0.3 \cdot u(0.8) + 0.64 \cdot u(1) + 0.06 \cdot u(4)$

$$\Leftrightarrow q \cdot 0.1 + (1-q) \cdot 1 = 0.3 \times 0 + 0.64 \times 0.1 + 0.06 \times 1$$

$$q = \frac{0.064 + 0.06 - 1}{0.9} = \frac{73}{79}.$$

So we only need $73/79 \approx 2.67\%$ chance of the better outcome now.

4. $L = [\alpha, 1-\alpha; w+2x, w-x]$

a) The agent's problem is to $\max_x EU[L(x)]$ s.t. $x \leq w$,

where $EU[L(x)] = \alpha \cdot u(w+2x) + (1-\alpha) \cdot u(w-x)$.

For which the FOC is $\frac{\partial EU}{\partial x} = 2\alpha \cdot u'(w+2x) + (\alpha-1) \cdot u'(w-x) \stackrel{!}{=} 0$

Informally: if we increase x , then - Since the individual's utility in money must be monotonically increasing so u' is everywhere -

holding x^* fixed we'd have both LHS terms increase. But for the FOC to be met with α' , their sum must remain zero.

By strict concavity of u (risk aversion), u' is decreasing in x . So if x^* fell when α rose, both $w+2x$ ^{gets smaller} and $w-x$ ^{gets smaller} and associated $u'(\cdot)$ get larger ^{smaller}. So FOC would fail to be met even more. $\oplus$ So we must have x^* increasing with α , so that it can offset the effect of α and FOC still holds.

More formally: use IFT.

$\oplus$ Since $\alpha-1 < 0$ and $2\alpha > 0$, these

both contribute to $\uparrow$ LHS. We let F be the LHS of FOC, which is C^1 for continuous u , has $F(\alpha, x^*) = 0$, and

satisfies $\left. \frac{\partial F}{\partial x} \right|_{x^*} = 4\alpha u''(w+2x) + (1-\alpha) u''(w-x) \leq 0 \neq 0$ by strict concavity.

So $\frac{\partial x^*}{\partial \alpha} = - \frac{\partial F / \partial \alpha}{\partial F / \partial x}$, where numerator is $2u'(\cdot) + u'(\cdot) > 0$ by increasing u .

and denominator < 0 , so $\frac{\partial x^*}{\partial \alpha} > 0$, as required.

b) We have $u(c) = \frac{c^{1-r}}{1-r}$; $u'(c) = c^{-r}$; $u''(c) = -r \cdot c^{-r-1}$

We want to solve $2\alpha \cdot (w+2x)^{-r} + (\alpha-1) \cdot (w-x)^{-r} = 0$ for x .

$$(2\alpha)^{-1/r} \cdot (w+2x) = (1-\alpha)^{-1/r} \cdot (w-x)$$

$$x \cdot (2(2\alpha)^{-1/r} + (1-\alpha)^{-1/r}) = w \cdot (-(2\alpha)^{-1/r} + (1-\alpha)^{-1/r})$$

$$\therefore x = w \cdot \frac{-\left(\frac{1-\alpha}{2\alpha}\right)^{1/r} + 1}{2\left(\frac{1-\alpha}{2\alpha}\right)^{1/r} + 1}$$

As r increases (higher relative risk aversion), the positive $\left(\frac{1-\alpha}{2x}\right)^{1/r}$ term ~~shrinks~~ ^{grows}, since $1/r$ smaller. For ~~increasing~~ ^{growing}

(Notice that $k := \frac{1-\alpha}{2x} = \frac{1}{2x} - \frac{1}{2} = \frac{1}{2}\left(\frac{1}{x} - 1\right)$. For the bet to be EU-positive (and hence EU-positive for any risk-averse agent, i.e. not $x^* = 0$) we need $x > \frac{1}{3} \Rightarrow k = 1$ and decreasing in x . So $k < 1$.)

Hence smaller numerator, larger denominator; x^* shrinks.

For log preferences we'd solve $\frac{2x}{w+2x} + \frac{\alpha-1}{w-x} = 0$ for x
 $\Rightarrow$ ~~area~~

$$2\alpha w - 2\alpha x = w + 2x - \alpha w - 2x\alpha$$

$$x^* = \frac{1}{2}w \cdot (3\alpha - 1)$$

Compare to the previous result, $x^* = w \cdot \frac{1 - \frac{1-\alpha}{2x}}{x^{\frac{1-\alpha}{2x}} + 1}$ with $r=2$

$$= w \cdot \frac{2\alpha - (1-\alpha)}{2(1-\alpha) + \alpha} = w \cdot \frac{3\alpha - 1}{3\alpha - 2} \quad \text{as expected.}$$

5. $f(z) = (\beta+1)z^\beta$ for $\beta \in [0, 1]$, over $z \in [0, 1]$

$$\begin{aligned} \text{a) } E[L(\beta)] &= \int_0^1 x f(x; \beta) dx = \int_0^1 x \cdot (\beta+1)x^\beta dx = \int_0^1 (\beta+1)x^{\beta+1} dx \\ &= \left[\frac{\beta+1}{\beta+2} x^{\beta+2} \right]_0^1 = \frac{\beta+1}{\beta+2} \end{aligned}$$

which increases in β by inspection,
or since it $= \frac{\beta+2-1}{\beta+2} = 1 - \frac{1}{\beta+2}$.

Lottery L_f FOSDs L_g iff $\int_0^x f(t) dt \leq \int_0^x g(t) dt \quad \forall x$, with strict $<$ for some

Notice $\int_0^x f(t; \beta) dt = x^{\beta+1}$ which for $x \in [0, 1]$ is decreasing in β .
So with larger β , $F_{\beta+1}(z) \leq F_{\beta-1}(z)$ everywhere, as required for FOSD.

This lets us conclude $\beta=2$ is the best lottery. The EV increasing in β didn't, because the tails could've got ^{heavier} in a way that risk-averse agents disliked. But all monotonic agents prefer FOSD lotteries, so we know $\beta=2$ is best.

b) $u(c) = c^{1-r}; \quad u'(c) = (1-r)c^{-r}.$

$$\begin{aligned} EU[L(\beta)] &= \int_0^1 u(z) \cdot f(z) dz = \int_0^1 z^{1-r} \cdot (\beta+1) z^\beta dz \\ &= \int_0^1 (\beta+1) z^{\beta+1-r} dz = \frac{\beta+1}{\beta+2-r}. \end{aligned}$$

$CE[L(\beta)]$ solves $EU[L(\beta)] = u(CE).$

So $CE = u^{-1}\left(\frac{\beta+1}{\beta+2-r}\right)$. Let $z^{1-r} = k$, then $z = k^{\frac{1}{1-r}}$.

$$= \left(\frac{\beta+1}{\beta+2-r}\right)^{\frac{1}{1-r}}$$

which we'd expect to decrease in r .

~~6a) $V(x) = \alpha u(x) + (1-\alpha)u(y)$~~

~~$V(x) = \alpha u(x) + (1-\alpha)u(y)$~~

~~$\alpha E[u(x)] + (1-\alpha)E[u(y)]$~~

~~The FOC for $\max_{\alpha} V(x)$ is $V'(\alpha) = 0$.~~

~~$V(\alpha) = E[\alpha u(x) + (1-\alpha)u(y)]$ by linearity.~~

~~so $V'(\alpha) = E[u(x) +$~~

6a) $V(\alpha) = \mathbb{E}[u(\alpha X + (1-\alpha)Y)]$.

The FOC for $\max V(\alpha)$ is $V'(\alpha) = 0$. By Leibniz's rule
 $\frac{\partial}{\partial \alpha} \mathbb{E}[\cdot] = \mathbb{E}\left[\frac{\partial}{\partial \alpha}(\cdot)\right]$.

So $V'(\alpha) = \mathbb{E}[(X-Y)u'(\alpha X + (1-\alpha)Y)]$ by chain rule, since
 $\frac{\partial}{\partial \alpha} u(\cdot) = (X-Y)u'(\cdot)$.

b) A risk-averse investor has positive but strictly decreasing u' , by concavity.

By linearity of $\mathbb{E}$, the FOC implies

$$\mathbb{E}[X u'(\alpha X + (1-\alpha)Y)] = \mathbb{E}[Y u'(\alpha X + (1-\alpha)Y)]$$

at optimum.

Suppose $\alpha = 0$. Then RHS is $\mathbb{E}[Y u'(Y)] < \mathbb{E}[Y] \mathbb{E}[u'(Y)]$
 by stated result.

and LHS is $\mathbb{E}[X u'(Y)]$.

$$= \mathbb{E}[X] \mathbb{E}[u'(Y)]$$

by assumption

But by independence, joint is product of marginals

$$= \mathbb{E}[X] \mathbb{E}[u'(Y)]$$

implying LHS > RHS

so FOC fails.

Similarly we can show it's a contradiction for $\alpha = 1$.

Thus $\alpha \in (0, 1)$.